

JISHNU JANARDANAN

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EDUCATION

Illinois Institute of Technology

B.S. in Computer Engineering

May 2026

Chicago, IL

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Systems Programming, Object-Oriented Programming, Database Organization, Computer Organization & Design, Machine Learning, VLSI Design, Probability and Statistics for Electrical and Computer Engineers

EXPERIENCE

Cloud Engineer (Project Manager) | Changing The Present | New York, NY, USA(Remote)

June 2026 – Present

- Analyzed a 23-table database in Python (pandas) and pgAdmin (PostgreSQL), found 31% of applicants silently dropped, 26% of tables dead.
- Moved the Postgres superuser password out of plain-text config into AWS Secrets Manager and gave each service a read-only role instead, with no downtime during the switch.
- Designed a BCNF schema and re-runnable ETL sync (Python, AWS Lambda, EventBridge, VPC) cutting data lag 99.9% (5 months to 1 hour).
- Led my colleagues through clear written briefs, delegating tasks via good communication, to build a Sheets-to-Postgres ETL pipeline.

Backend Developer Intern | FoodCLUB | London, United Kingdom (Remote)

Jan 2024 – May 2024

- Held authentication and data APIs at 99% uptime by building and maintaining Node.js services on AWS EC2
- Cut response latency 30% on the busiest endpoints by adding Redis caching, making the application faster

CERTIFICATIONS & PUBLICATIONS

Certifications: NVIDIA-Certified Professional: Agentic AI (NCP-AAI) ([Link](#)) | Building RAG Agents with LLMs ([Link](#)) | Building Agentic AI Applications with Large Language Models ([Link](#)) | Claude Code 101 ([Link](#)) | AI Fluency: Framework & Foundations ([Link](#)) | Introduction to Model Context Protocol ([Link](#))

Publication: ThinkFast: an embedded trivia game (Paper ID 121, under review, IEEE EIT 2026)

TECHNICAL SKILLS

Programming Languages – Python, Java, JavaScript, HTML/CSS, SQL, Bash, C++

Frameworks & Libraries – React, Vue 3, Vite, FastAPI, LangGraph, LangChain, Electron, Flutter, Node.js, Spring Boot, pandas, JUnit

AI/ML & LLM Systems – LLM Routing, LLM-as-a-Judge, Retrieval-Augmented Generation, Vector Search (FAISS, sentence-transformers), LoRA Fine-Tuning (Hugging Face PEFT), LLM Evaluation (RAGAS), NeMo Guardrails, LangSmith Tracing, Agent Loops (LangGraph), Model Context Protocol (MCP), Multi-provider LLM APIs (Anthropic, OpenAI, Gemini, NVIDIA NIM)

Cloud & DevOps – AWS (Lambda, EC2, VPC, ECR, ECS Fargate, API Gateway, RDS, S3, CloudFront, SES, EventBridge, IAM, Secrets Manager, SSM, Route 53, CloudFormation, CloudShell, CloudWatch, Cost Explorer), GCP (Firebase Auth, YouTube Data API, Google Colab (T4), Terraform, Docker, Kubernetes (kind, HPA), GitHub Actions CI/CD, Prometheus/Grafana, Caddy, Server-Sent Events, MQTT, Linux/Ubuntu)

Databases & System Design – PostgreSQL, pgAdmin (PostgreSQL), MySQL, DynamoDB, Redis, Firebase - Lock-Free Concurrency (MPSC/SPSC), Memory Pool Allocation, REST, JWT, Rate Limiting, Idempotency, Pub/Sub, TCP, UDP Multicast, Multi-Threading

PROJECTS

slice – AI & Cloud Cost Gateway | Python, FastAPI, LangGraph, PostgreSQL, Redis, FAISS, Terraform, Docker, Kubernetes, AWS (EC2, ECR, S3, Secrets Manager, SSM, IAM, Route 53, CloudFormation, Cost Explorer) | [GitHub](#) | <https://sliceapp.dev/> Summer 2026

- Cut AI spend 45% on a 50-prompt workload by routing 54% of prompts to a cheaper model and caching repeats in Redis
- Built LLM routing (LangGraph, pin > rule > auto) with a try-cheap, escalate-on-fail loop capped at 3 tries, \$0.25/request, 47 tests
- Put four providers (Anthropic, OpenAI, Gemini, NVIDIA NIM) behind one gateway endpoint plus an OpenAI-compatible twin, via one env var
- Hit 92% routing accuracy (54% baseline) by LoRA fine-tuning Qwen2.5-0.5B on 654 routing decisions in 75 seconds on a free Google Colab T4
- Confirmed cheap answers held up with an LLM-as-a-judge (RAGAS, 0.89 relevancy), keeping 99.9% of the live router's quality
- Guarded the agent with NeMo Guardrails (in/out rails, fail-open, 5s) via LangChain, traced LangGraph steps in LangSmith, 25 tests
- Survived Redis, Postgres, and vector-search outages by making every non-auth path fail open, covered by 608 unit tests
- Isolated each account's spend, cache, and budget with GitHub device-flow login, SHA-256 hashed API keys, and JWT sessions
- Ran it in production on arm64 EC2 with Terraform, Prometheus/Grafana, and HPA autoscaling 2 to 4 pods verified on a local kind cluster
- Exposed spend, budgets, and routing rules inside Claude Code via a 7-tool MCP server where reads are free and writes confirm
- Built a Mac desktop app with Electron, so users run slice in a window instead of the terminal
- Built a read-only boto3 scanner flagging public S3 buckets, open ports, and idle instances via a one-click CloudFormation role

C++ Financial Trading Exchange | C++26, Bazel, GoogleTest, BUILD Fellowship | [GitHub](#) | [Certificate](#)

Summer 2026

- Built an O(1) limit order book with price-time priority matching, backed by ordered price maps and an order-id iterator index
- Cut client latency to a single enqueue with a fixed-size memory pool and a lock-free MPSC ring queue off the hot path
- Verified 4 clients rebuild an 8,235-order book bit-for-bit across 111,842 messages with 96.6% line / 94.9% branch coverage
- Modeled exchange connectivity with TCP for order entry and UDP multicast for market-data, per the Nasdaq ITCH/OUCH model
- Mentored by a tech lead from Citadel

SneakersForLess – Sneaker Price Comparison Site | React, AWS Lambda, DynamoDB, SES, Firebase | [GitHub](#) | <https://www.sneakersforless.org/> Spring 2026

- Surfaced the lowest sneaker price across 25+ retailers with a ranking engine, weighted relevance search, and watchlists
- Secured user data by verifying Firebase tokens on the serverless backend, with a least-privilege IAM role per Lambda
- Cut deploy time 80% with GitHub Actions CI/CD to S3 and CloudFront, and stopped duplicate alerts with an idempotency stamp
- Automated price-drop emails via AWS SES, alerting watchlist users the moment a tracked sneaker hit a new low

Old Is Gold, Senior Fitness Platform | JavaScript, AWS Lambda, DynamoDB | [GitHub](#) | <https://oldisgold.fit/>

Spring 2026

- Delivered workout plans and progress tracking for users 55+, keeping the billable YouTube API key behind a server-side proxy
- Blocked bad tokens and abusive traffic with fail-closed RS256 JWT checks and fail-open per-user DynamoDB rate limiting
- Held responses under 200ms by keying all data to user_id in one DynamoDB partition, making each dashboard load a single query

IoT Smart Device Hub | Java, Spring Boot, MQTT, JUnit, Linux/Ubuntu | Illinois Institute of Technology

Spring 2025

- Synced device state in real time across HTTP and MQTT with a Spring Boot backend on pub/sub and the observer pattern
- Reached 94% instruction coverage and a 100% pass rate (JaCoCo) with a JUnit suite of 30+ tests